Frequency-Corrected Proportional Navigation

Problem Statement

- Proportional navigation with third-order flight control
- Sinusoidal (weaving) target maneuvers
- Miss distance strongly depends on maneuver frequency

Flight-Control Model

$$W(s) = \frac{1 - s^2/\omega_z^2}{(1 + \tau s)\left(1 + \frac{2\zeta}{\omega_M}s + \frac{s^2}{\omega_M^2}\right)} \tag{1}$$

- τ: time constant
- ζ: damping ratio
- ω_M: natural frequency

Miss-Distance Frequency Response

$$|P(i\omega)| = \frac{1}{\omega^2}\left(\omega^2 + \frac{1}{\tau^2}\right)^{-\frac{NB}{2\tau}} \left[(\omega_M^2 - \omega^2)^2 + 4\zeta^2\omega_M^2\omega^2\right]^{-\frac{NC}{2}} \tag{2}$$

Most Dangerous Frequency

$$\omega^* \approx \omega_M \sqrt{1 - 2\zeta^2} \tag{3}$$

- Peak miss distance near ω^*
- Strong dependence on damping ζ

Frequency-Corrected PN

$$N_{\text{eff}}(\omega) = N_0 \left[\frac{(\omega_M^2 - \omega^2)^2 + 4\zeta^2\omega_M^2\omega^2}{\omega_M^4} \right]^\gamma \tag{4}$$

$$\hat{\omega}^2 = \left| \frac{\ddot{\lambda}}{\dot{\lambda}} \right| \tag{5}$$

Selection of γ

- γ controls gain reduction near ω^*
- $\gamma \in [0.25, 0.5]$ recommended
- $\gamma = 0.3$ used in simulations

Simulation Results (One-Slide Summary)

- Three cases compared: $\gamma = 0$ (PN), $\gamma = 0.3$, $\gamma = 0.5$
- Target executes sinusoidal maneuver near ω^*
- Miss distance $y(t_f)$ reduced significantly with frequency correction
- $\gamma = 0.3$: best trade-off between miss reduction and smooth control
- Results consistent with frequency-domain prediction of $|P(i\omega)|$

Conclusion

- Frequency correction reduces peak miss distance
- No additional phase lag introduced
- Suitable for real-time implementation